

ABC Phones

Data Engineering & Analytics Case Study

ETL Pipeline · Data Quality · Portfolio Analysis

Python · Pandas · Pandera · PostgreSQL · Docker

ETL Pipeline Architecture

How data moves from source to insight



Full load strategy — no CDC markers in source systems

Key Engineering Decisions

Six choices that shaped the pipeline

01	Ingestion strategy	Full load for all sources — credit snapshots are point-in-time, sales has no CDC markers, NPS refreshes monthly.
02	Column standardisation	standardize_columns() runs on every load: lowercase, underscores, consistent across all three datasets.
03	Extra column detection	Unnamed: 28 appeared in Jun/Sep/Dec snapshots. Pandera schema validation auto-detects and alerts on column drift.
04	DOB timezone handling	Mixed timezones parsed with utc=True then made tz-naive. Age is calculated relative to each snapshot date.
05	Join strategy	Sales + DOB → customer master (5.4% match rate). Gender/Income NaN loan IDs excluded. NPS joins on loan_id directly.
06	Storage design	PostgreSQL partitioned by snapshot_date across raw → staging → analytics schemas. CSV exports for stakeholders.

Data Quality Framework

Five checks, one real example

Check	What it does	Severity	Frequency
Freshness	5 snapshots expected — alerts if any are missing	HIGH	Per batch
Uniqueness	No duplicate (loan_id, date) pairs	HIGH	Daily
Referential integrity	All loan_ids must exist in customer master	CRITICAL	Daily
Range / plausibility	Ages 18–120, balance ≥ 0 and other bounds	MEDIUM	Weekly
Null thresholds	% nulls per column kept within defined tolerance	MEDIUM	Daily

Real example —

An extra unnamed column appeared in three snapshots (Jun, Sep, Dec). Pandera caught the mismatch, fired a webhook to Slack and PagerDuty, and escalated to the Data Engineer → Lead → Head of Data until resolved.

Portfolio Health — Key Metrics

Measured across 5 monthly snapshots, Jan – Dec 2025

41.9% → 45.3%

Delinquency rate

13.5% → 17.7%

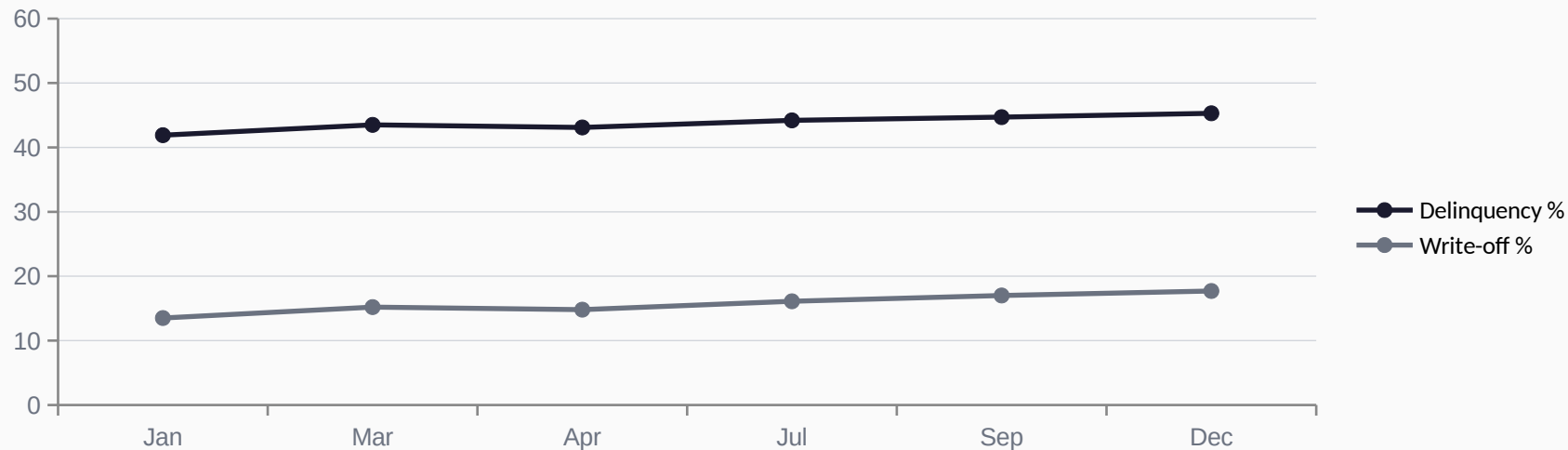
Write-off rate

75.4% → 72.7%

Collection rate

170 → 273 days

Avg days past due



Who Carries the Risk?

Critical risk varies significantly across age and income bands

By age band

18–25 years old: 22.4% Critical (highest of any group)

26–35 years old: 18.1% Critical

36–45 years old: 15.9% Critical

46–55 years old: 12.6% Critical (lowest)

55+ years old: 13.8% Critical

By income band

Below 5,000: 27.3% Critical (highest)

5,000–15,000: 21.4% Critical

15,000–50,000: 18.0% Critical

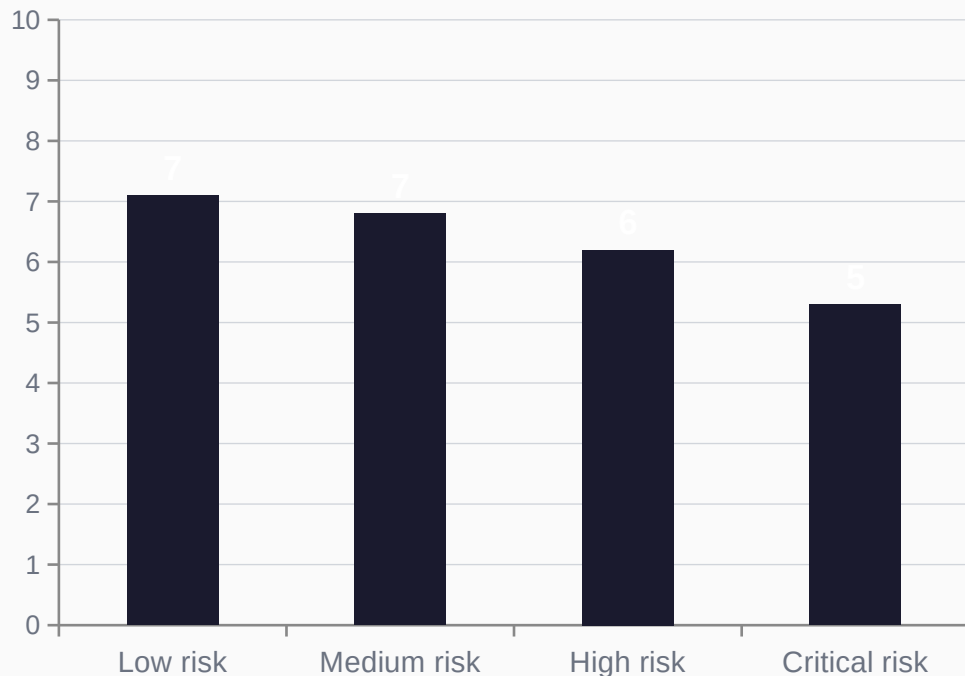
50,000–150K: 15.6% Critical

150,000+: 14.4% Critical (lowest)

Younger and lower-income customers carry disproportionately more risk. Recommendation: stricter initial credit limits for the 18–25 cohort.

Credit Risk and Customer Satisfaction

Customers in arrears score lower — but fair collections preserve NPS



5.3

NPS for critical risk customers
(Detractors)

7.1

NPS for low risk customers
(Passive)

4.6

NPS when 365+ days past due

Customers who received >100% collection rate still averaged NPS 7.1 — respectful, proactive engagement preserves satisfaction.

Recommendations

What we'd change and what data we'd add

What to do now

1. Launch a Payment Difficulty programme for customers who enter arrears at 7–30 days DPD — proactive outreach before accounts reach 90 days.
2. Split collections into two tracks: a soft customer-success track for early-stage accounts, and a hard legal track only for late-stage.
3. Apply stricter initial credit limits to the 18–25 cohort, given their 22% critical risk rate.

Data we wish we had

- Standardise loan_id naming across all systems with foreign key constraints
- Replace the four ad-hoc income columns with a structured (loan_id, income_type, amount, verified_date) table
- Add a controlled vocabulary for account_status_l1 with defined transition rules
- Include region/city for geographic concentration risk monitoring
- Track device return reason codes for loss-given-default modelling

Summary

1. Built an end-to-end ETL pipeline processing 71K+ credit records across 5 snapshots.
2. Implemented 5 data quality checks with Pandera schema validation and webhook alerting.
3. Delinquency has risen to 45.3% with write-offs reaching 17.7% — the portfolio is under pressure.
4. The 18–25 age band and sub-5K income bracket carry the highest critical risk.
5. There is a clear inverse relationship between credit risk and NPS — but fair collections protect it.
6. Recommended: early intervention at 7–30 days DPD, two-track collections, stricter young-cohort limits.